

Computing the Centre of Elasticity in Planar Frames

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1. Introduction

1.1 On spectral decomposition of stiffness matrix

2. Rigid-body transformation of point of interest

3. Serial concatenation of elements

4. Parallel concatenation of elements

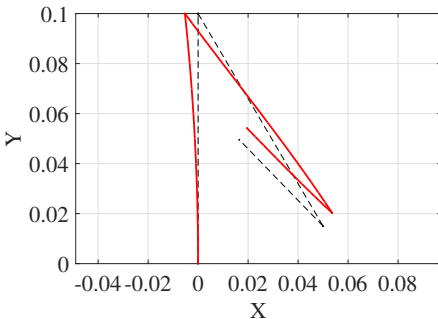
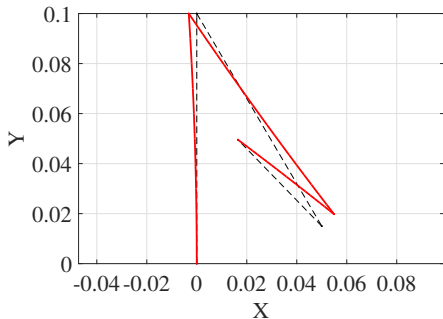
5. Conclusions

Center of Elasticity



At the **center of elasticity** (CoE), the rotational and translational degrees of freedom are decoupled,

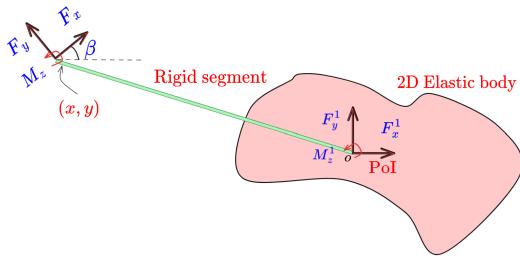
- linear forces applied at this point only cause translational displacement,
- moment applied at this point causes only rotational displacement.



Center of Elasticity



- The CoE **does not refer to any material point in a continuum body**.
- The CoE **maps each material point to a particular spatial coordinate**.
- If this point were to be connected with a **rigid link**, the translation and rotation about this particular spatial coordinate would be decoupled.
- Existence of CoE is **guaranteed in planar bodies**.





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Let us consider a solid (isotropic linear elastic) body subjected to transverse loads and moments at some point(s) in the body; then the displacement (translational and rotational) at some other point(s) can be represented as:

$$\{f\} = [K]\{u\}$$

In mechanics, **normal forms of symmetric tensors are usually preferred for analysis and interpretation.**

Stiffness matrix (contd..)



Consider the stiffness matrix for a cantilever beam:

$$[K] = \begin{bmatrix} EA/L & 0 & 0 \\ 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & -6EI/L^2 & 4EI/L \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & c & d \end{bmatrix}$$
$$[v_1, v_2, v_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2L}{3} - \frac{\sigma_2 + \sigma_1 + EI L^2}{L \sigma_2} & \frac{2L}{3} - \frac{\sigma_2 - \sigma_1 + EI L^2}{L \sigma_2} \\ 0 & 1 & 1 \end{bmatrix} \text{ and}$$
$$D = \begin{bmatrix} \frac{AE}{L} & 0 & 0 \\ 0 & \frac{2(3EI + \sigma_1 + EI L^2)}{L^3} & 0 \\ 0 & 0 & \frac{2(3EI - \sigma_1 + EI L^2)}{L^3} \end{bmatrix}$$

where, $\sigma_1 = EI\sqrt{L^4 + 3L^2 + 9}$ * and $\sigma_2 = 3EI$.

*Notice the incompatibility of dimensions!

Stiffness matrix (contd..)



As is usually done, let us attempt to do a rigid body transformation (only rotate the basis) to diagonalise the matrix.

Then

$$\mathbf{Q} = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$$

but this is not allowed!!!

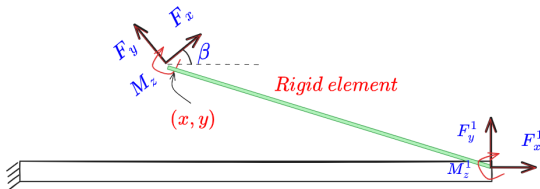
As for frame elements, admissible $\mathbf{Q}$ for rigid body rotation in this case is :

$$\mathbf{Q} = \begin{bmatrix} I & m & 0 \\ -m & I & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Stiffness matrix (contd..)



Thus, just rigid body rotation alone won't suffice here. We also have to perform rigid-body translation of our point of interest.

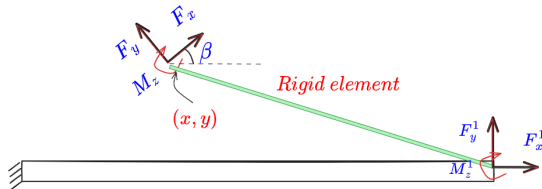


$$F_y^1 = F_y \cos \beta + F_x \sin \beta$$

$$F_x^1 = F_x \cos \beta - F_y \sin \beta$$

$$M_z^1 = M_z - F_y(L - x) - F_x y$$

Stiffness matrix under RB transformation (contd..)



$$\begin{Bmatrix} F_x^1 \\ F_y^1 \\ M_z^1 \end{Bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ -y \cos \beta - (L-x) \sin \beta & -(L-x) \cos \beta + y \sin \beta & 1 \end{bmatrix} \begin{Bmatrix} F_x \\ F_y \\ M_z \end{Bmatrix},$$

$$\begin{Bmatrix} u_x^1 \\ v_y^1 \\ \theta_z^1 \end{Bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta & l^* \sin \delta \\ \sin \beta & \cos \beta & l^* \cos \delta \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_x \\ v_y \\ \theta_z \end{Bmatrix},$$

where $\delta = \tan^{-1}(\frac{y}{L-x})$ and $l^* = \sqrt{(L-x)^2 + y^2}$.

Stiffness matrix under RB transformation (contd..)



$$\{\mathbf{f}^1\} = [\mathbf{K}]\{\mathbf{u}^1\}$$

$$\{\mathbf{f}\} = [\mathbf{K}']\{\mathbf{u}\}$$

where

$$[\mathbf{K}'] = \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ -y \cos \beta - (L-x) \sin \beta & -(L-x) \cos \beta + y \sin \beta & 1 \end{bmatrix}^{-1} [\mathbf{K}] \begin{bmatrix} \cos \beta & -\sin \beta & l^* \sin \delta \\ \sin \beta & \cos \beta & l^* \cos \delta \\ 0 & 0 & 1 \end{bmatrix}$$

Stiffness matrix under RB transformation (contd..)

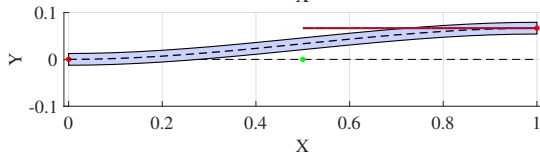
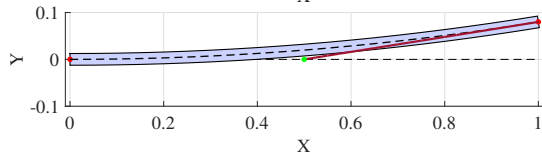
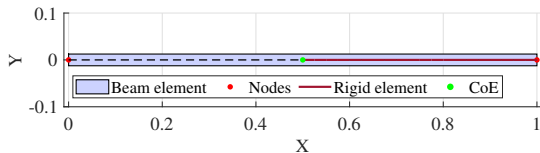
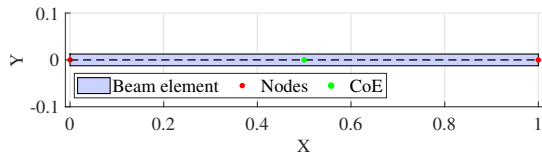


Let us consider the case $x = L/2$, $y = 0$, $\beta = 0$: where

$$\begin{aligned} [K'] &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -L/2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} EA/L & 0 & 0 \\ 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & L/2 \\ 0 & 0 & 1 \end{bmatrix} \\ [K'] &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & L/2 & 1 \end{bmatrix} \begin{bmatrix} EA/L & 0 & 0 \\ 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & L/2 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 0 \\ 0 & 0 & EI/L \end{bmatrix} \end{aligned}$$

The stiffness matrix is diagonal!

CoE of Cantilever beam about its tip





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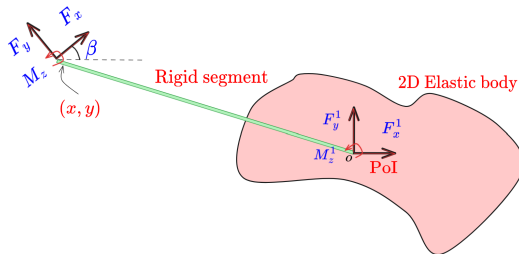
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Rigid-body transformation of point of interest



$$\begin{Bmatrix} F_x^1 \\ F_y^1 \\ M_z^1 \end{Bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ -y \cos \beta + x \sin \beta & x \cos \beta + y \sin \beta & 1 \end{bmatrix} \begin{Bmatrix} F_x \\ F_y \\ M_z \end{Bmatrix},$$
$$\begin{Bmatrix} u_x^1 \\ v_y^1 \\ \theta_z^1 \end{Bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta & y \\ \sin \beta & \cos \beta & -x \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_x \\ v_y \\ \theta_z \end{Bmatrix}$$

Rigid-body transformation of point of interest



$$[K] = \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ -y \cos \beta + x \sin \beta & x \cos \beta + y \sin \beta & 1 \end{bmatrix}^{-1} [K^1] \begin{bmatrix} \cos \beta & -\sin \beta & y \\ \sin \beta & \cos \beta & -x \\ 0 & 0 & 1 \end{bmatrix}$$

Rigid-body translation of Pol



$$[K] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y & x & 1 \end{bmatrix}^{-1} [K^1] \begin{bmatrix} 1 & 0 & y \\ 0 & 1 & -x \\ 0 & 0 & 1 \end{bmatrix} =$$
$$\begin{bmatrix} K_{11}^1 & K_{12}^1 & K_{31}^1 - x(K_{12}^1) + y(K_{11}^1) \\ \cdot & K_{22}^1 & K_{23}^1 - x(K_{22}^1) + y(K_{12}^1) \\ \cdot & \cdot & K_{33}^1 + (K_{11}^1)y^2 + (K_{22}^1)x^2 - (2K_{12}^1)xy + (2K_{13}^1)y - (2K_{23}^1)x \end{bmatrix}$$

Rigid-body translation of Pol



$$[\mathbf{C}] = \begin{bmatrix} 1 & 0 & y \\ 0 & 1 & -x \\ 0 & 0 & 1 \end{bmatrix}^{-1} [\mathbf{C}^1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y & x & 1 \end{bmatrix} =$$
$$\begin{bmatrix} C_{11}^1 - (2C_{13}^1)y + C_{33}^1y^2 & (C_{33}^1)x^2 + (2C_{23}^1)x + C_{22}^1 & \cdot \\ C_{12}^1 + (C_{13}^1)x - (C_{23}^1)y - (C_{33}^1)xy & C_{23}^1 + (C_{33}^1)x & \cdot \\ C_{13}^1 - (C_{33}^1)y & & C_{33}^1 \end{bmatrix}$$

Rigid-body rotation of Pol



$$\begin{aligned} [\mathbf{K}] &= \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} [\mathbf{K}^1] \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} K_{11} & K_{12} & K_{13}^1 \cos(\beta) + K_{23}^1 \sin(\beta) \\ \cdot & K_{22} & K_{23}^1 \cos(\beta) - K_{13}^1 \sin(\beta) \\ \cdot & \cdot & K_{33}^1 \end{bmatrix} \end{aligned}$$

Where,

$$K_{11} = K_{11}^1 \cos^2(\beta) + K_{22}^1 \sin^2(\beta) + K_{12}^1 \sin(2\beta)$$

$$K_{12} = K_{12}^1 \cos(2\beta) + 0.5(K_{22}^1 - K_{11}^1) \sin(2\beta)$$

$$K_{22} = K_{22}^1 \cos^2(\beta) + K_{11}^1 \sin^2(\beta) - K_{12}^1 \sin(2\beta)$$



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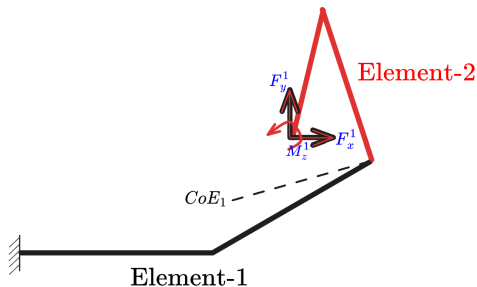
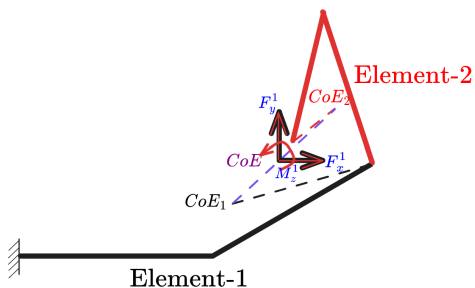
Serial concatenation of elements



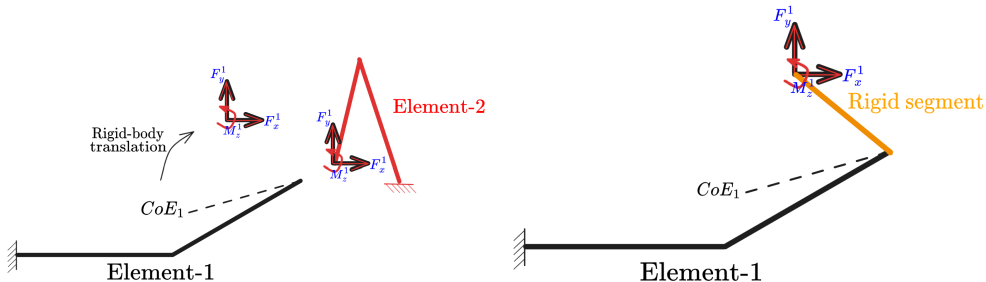
Let the single-point stiffness matrix of **element-1** about $CoE_1 (\{x_1, y_1\})$ and **element-2** about $CoE_2 (\{x_2, y_2\})$ be given by:

$$[C_1] = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{22} & 0 \\ 0 & 0 & C_{33} \end{bmatrix},$$

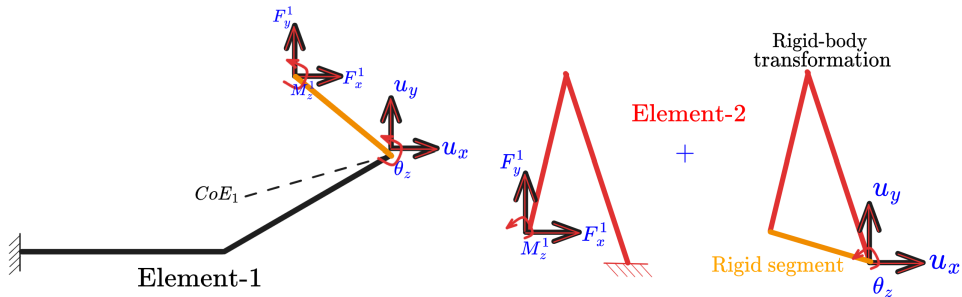
$$[C_2] = \begin{bmatrix} G_{11} & G_{12} & 0 \\ G_{12} & G_{22} & 0 \\ 0 & 0 & G_{33} \end{bmatrix}$$



Serial concatenation of elements



Serial concatenation of elements



Serial concatenation of elements



$$[\mathbf{C}_t^{net}] = \begin{bmatrix} C_{11} + G_{11} & C_{12} + G_{12} & -C_{33}(y_t - y_1) - G_{33}(y_t - y_2) \\ . & C_{22} + G_{22} & C_{33}(x_t - x_1) + G_{33}(x_t - x_2) \\ . & . & C_{33} + G_{33} \end{bmatrix}$$

$$(x_{CoE}, y_{CoE}) = \left(\frac{(C_{33})x_1 + (G_{33})x_2}{C_{33} + G_{33}}, \frac{(C_{33})y_1 + (G_{33})y_2}{C_{33} + G_{33}} \right)$$

Serial concatenation of beam elements

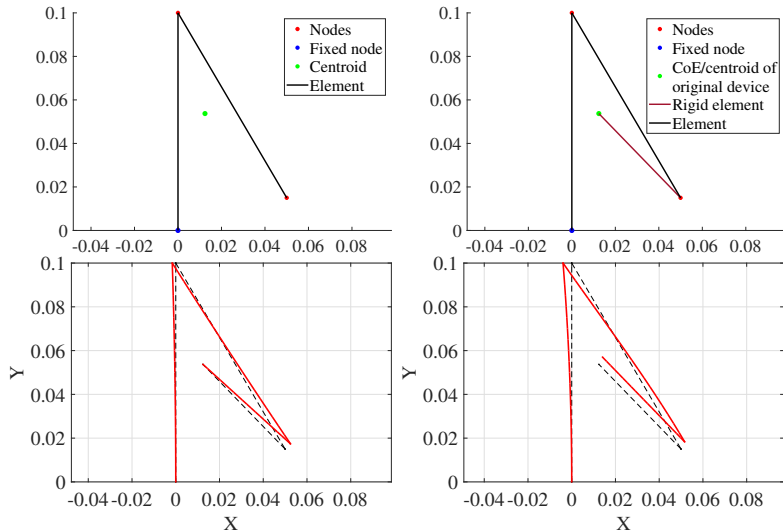


For beam elements: $C_{33}|_{\text{CoE}} = L/(EI)$

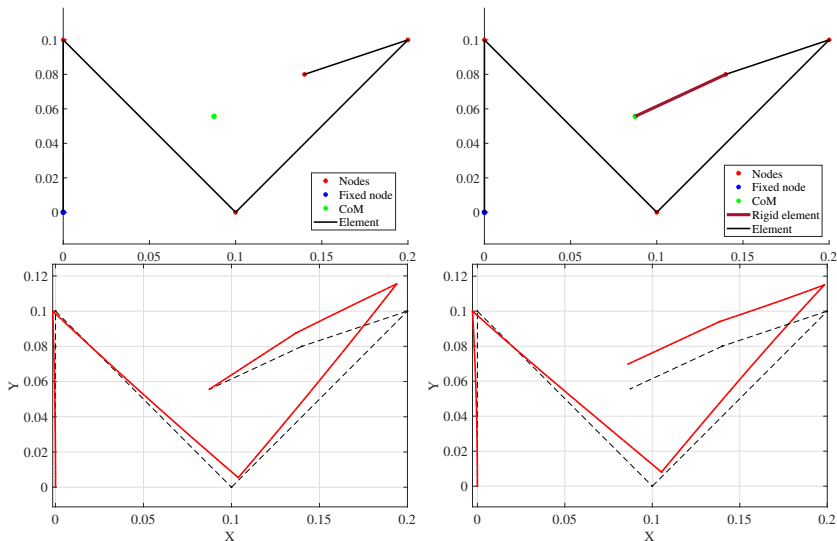
$$\Rightarrow \boxed{\mathbf{r}_{\text{CoE}} = \frac{\sum_{i=1}^n (L_i/E_i I_i) \mathbf{r}_{m,i}}{\sum_{i=1}^n (L_i/E_i I_i)}}$$

Thus, for elements with uniform flexural rigidity, the **CoE and centroid** (in skeletal frame representation) **are coincident**.

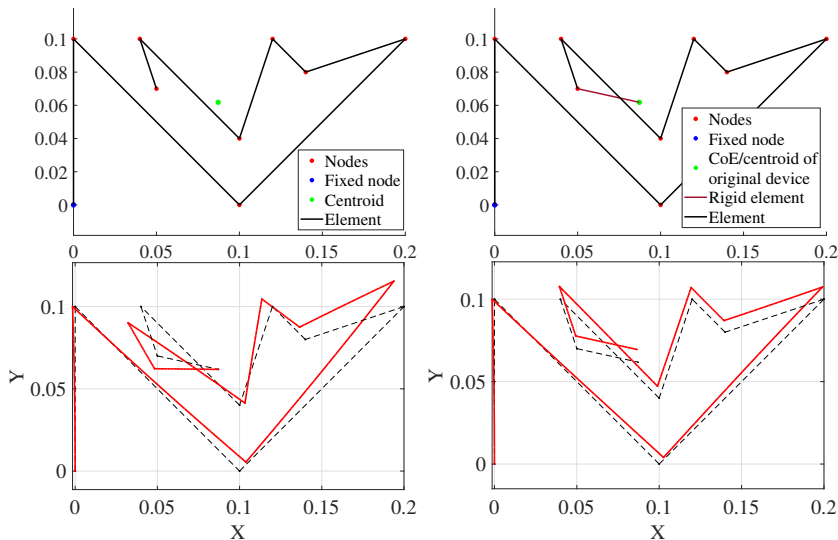
Serial concatenation of beam elements- Examples



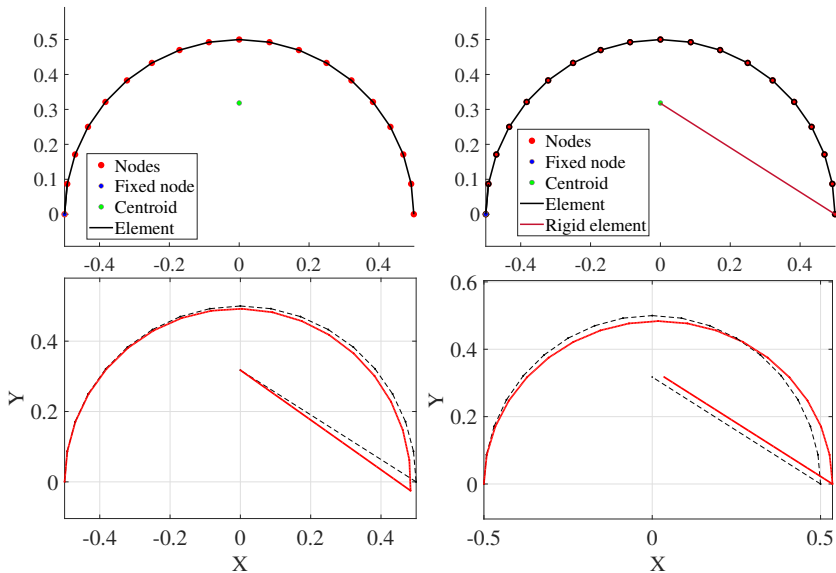
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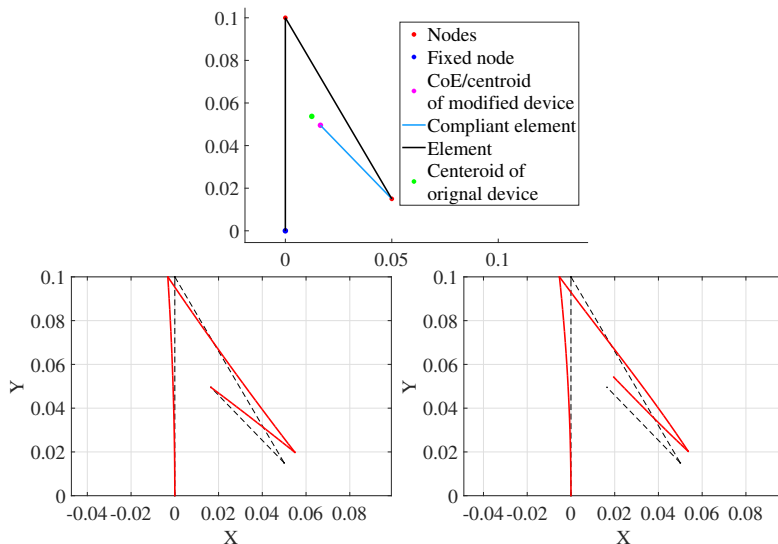
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Serial concatenation of beam elements- Examples





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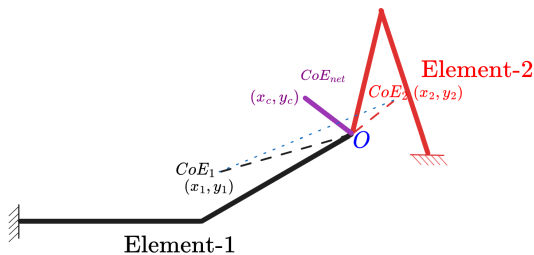
Parallel concatenation of elements



$$[K_1] = \begin{bmatrix} K_{11} & K_{12} & 0 \\ K_{12} & K_{22} & 0 \\ 0 & 0 & K_{33} \end{bmatrix}$$

$$[K_2] = \begin{bmatrix} P_{11} & P_{12} & 0 \\ P_{12} & P_{22} & 0 \\ 0 & 0 & P_{33} \end{bmatrix}$$

$$\begin{bmatrix} K_{11} + P_{11} & K_{12} + P_{12} \\ K_{12} + P_{12} & K_{22} + P_{22} \end{bmatrix} \begin{Bmatrix} y_{CoE} \\ -x_{CoE} \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{bmatrix} \begin{Bmatrix} y_1 \\ -x_1 \end{Bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \begin{Bmatrix} y_2 \\ -x_2 \end{Bmatrix}$$





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Towards systematic synthesis of planar flexure joints



Effective planar flexure joints:

- Connected to CoE.
- Allow for large deformation.
- Be stiff against undesired deformation modes.

$$[K'] = \begin{bmatrix} k_{xx} & 0 & 0 \\ 0 & k_{yy} & 0 \\ 0 & 0 & k_{\theta\theta} \end{bmatrix}$$

Center of Elasticity (3D)



In 2D, 3 variables are needed to define a point (x, y, β) . But how many equations do we have?

$$\{\mathbf{f}\}_{3 \times 1} = [\mathbf{K}']_{3 \times 3} \{\mathbf{u}\}_{3 \times 1}$$

- **Case-1:** $f_x = 0, f_y = 0, m_z \implies u_x = 0, u_y = 0, \theta_z$ (2 equations)
- **Case-2:** $f_x = 0, f_y, m_z = 0 \implies u_x = 0, u_y, \theta_z = 0$ (2 equations)
- **Case-3:** $f_x, f_y = 0, m_z = 0 \implies u_x, u_y = 0, \theta_z = 0$ (2 equations)

We have six equations, of which 3 are degenerate (consequence of *Maxwell's Reciprocal theorem*). Thus, we have three equations and three unknowns; here the equations are nonlinear. But the existence of a solution is guaranteed.

Other kinds of elastic centers



In 3d, consider the following stiffness matrix:

$$[\mathbf{K}]_{6 \times 6} = \begin{bmatrix} \mathbf{A}_{3 \times 3} & \mathbf{B}_{3 \times 3} \\ \mathbf{B}_{3 \times 3}^T & \mathbf{C}_{3 \times 3} \end{bmatrix},$$

where $\mathbf{A}$ and $\mathbf{C}$ are symmetric.

If we can perform a rigid body transformation such that the off-diagonal matrix ($\mathbf{B}$) in the new coordinate frame is symmetric (or equivalently diagonal), then that point is defined as *centre of stiffness*.

At this point, the rotation and translation components are said to be maximally decoupled[†] (if $\mathbf{B}$ can be made zero, this point is its centre of elasticity).

[†]J. Loncaric. “Normal forms of stiffness and compliance matrices”. In: *IEEE Journal on Robotics and Automation* 3.6 (1987), pp. 567–572. DOI: 10.1109/JRA.1987.1087148.



- The concept of the CoE is elucidated through the **stiffness and compliance matrix**.
- Relations for finding the CoE of devices obtained through **serial and parallel concatenation of elements** are shown here.
- For a serial concatenation of beam elements of uniform flexural rigidity, the CoE of the device about the terminal node is **coincident with the centroid** of the device (in its skeletal frame representation).



Thank you!